AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 7 · STUDENT LESSON

Categories and Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 4: a rule must specify how unfamiliar cases are handled.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Apply a category definition using explicit evidence and boundary rules. Try the recall before opening notes.

Your learning target

I can apply a **category** definition using explicit **evidence** and boundary rules.

I can distinguish a known failure from information that is missing or unreadable.

I can use **disagreement** to improve a **label**ling guide without mistaking agreement for truth.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Labels are human choices that may **overlap** or be unclear.



Your target

LEARNING TARGET Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

Learner name:

Readiness for the connected explanation

A transcript leaves out an unclear word. Why should a later system not simply treat that missing word as a known value?

This week's end goal

By the end, I can apply a shared **label**ling policy consistently, including missing information.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Categories and Labels case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A playground object can be both red and round, a toy and sports equipment, or safe in one situation and unsafe in another.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Two students are sorting fictional event cards for a classroom calendar. Their first instruction is Put the good cards together. Mina calls a colourful card good. Theo calls a plain card good because its room is easy to read.

They disagree even though both followed a reasonable interpretation of the word good.

A **category** is useful for a task when people can tell what **evidence** makes an example belong to it. An **operational definition** states the **observation**s or checks to perform.

For this paper exercise, we will check two printed fields: an event day and a room code. The task is to check the record against a classroom policy, not to judge the quality of the event or the person who wrote it.

Labels depend on the definition in use. A model fitted to **label**s can learn the distinctions those **label**s encode, including mistakes and inconsistencies.

Before asking a model to copy a **label**, ask whether people have a clear, useful guide for assigning it.

STUDENT LESSON · 3 OF 18

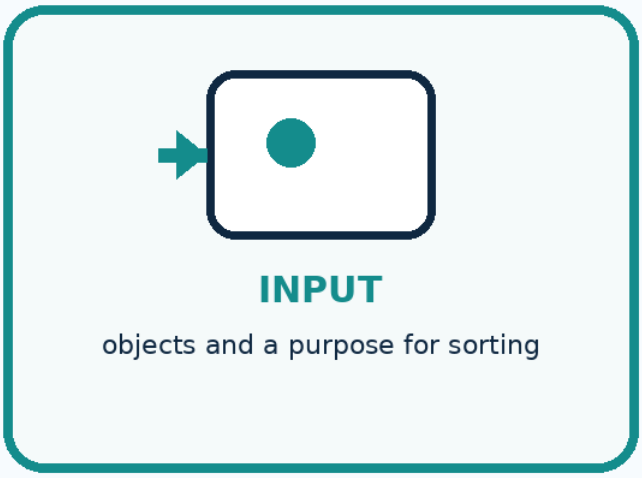
See how it works

> Follow the mechanism for Categories and Labels. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

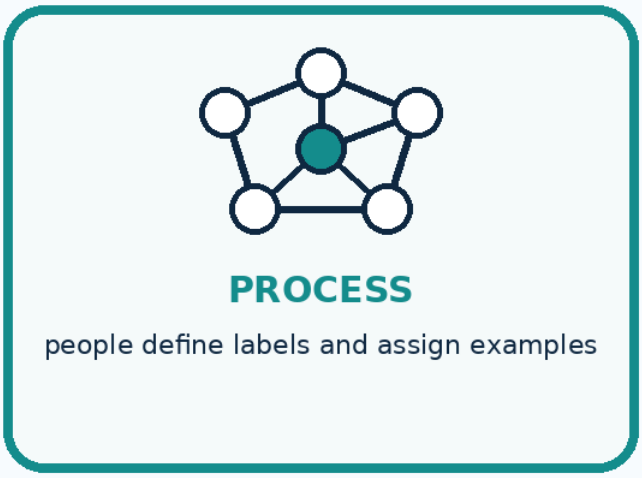
Input → Process → Output

Follow the information instead of imagining magic inside the machine.



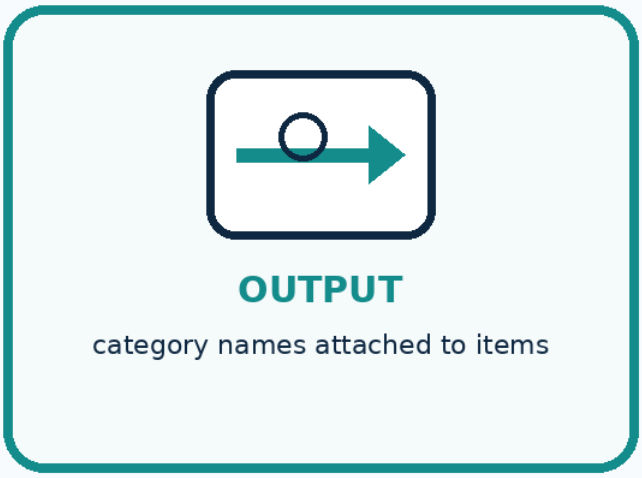
INPUT

objects and a purpose for sorting



PROCESS

people define labels and assign examples



INPUT objects and a purpose for sorting

PROCESS people define **label**s and assign examples

OUTPUT category names attached to items

Key idea

CORE IDEA **Label**s are human choices that may **overlap** or be unclear.

How the mechanism works

UNDER THE HOOD A **label** is an **operational definition** chosen for a task. Categories may be mutually exclusive, multi-**label**, hierarchical, subjective, or context-dependent.

Ambiguity in **label**s becomes **ambiguity** in **training** and evaluation.

Draw a fresh example with all three stages:

A closer explanation

Our fictional calendar accepts a whole-number day from 1 through 30, including both endpoints, and exactly one room code, A or B. The month is already fixed for this exercise. A blank or unreadable field counts as unknown.

The categories are revise, review and ready. Use the checks in the order shown.

Three ordered checks for a calendar card

CHECK IN THIS ORDER	EVIDENCE	RECORDED LABEL
First check known values	Any known field fails its criterion	Revise even if another field is unknown
Then check missing information	No known failure but a field is blank or unreadable	Review because evidence is incomplete
Otherwise all checks have passed	Day is 1 through 30 and room is A or B	Ready under this record policy

Stop at the first matching row. The order resolves a card with both a known failure and an unknown field. Ready describes the printed record under this policy; it does not confirm that the event exists or has permission to run.

Known failure and unknown are different. Day 31 is known and outside the accepted range, so the card needs revision. An unreadable day might be valid or invalid; without another known failure, it needs review.

Never silently turn a blank into zero just to make a comparison possible.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

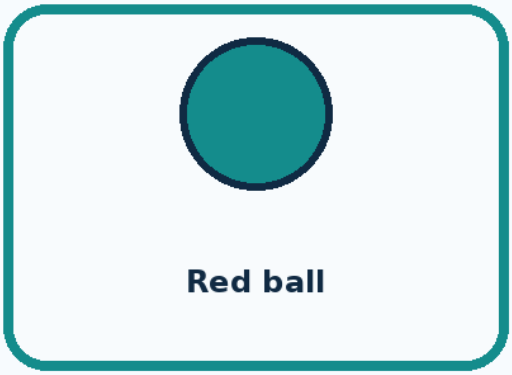
Compare the cases

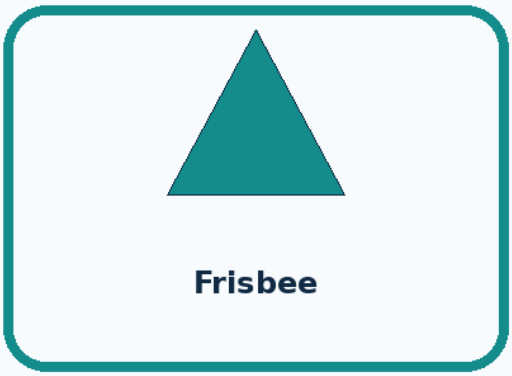
> Compare the supplied cases for Categories and Labels. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Red ball Clue: red and round. Meaning: multiple descriptive **label**s.

The case to investigate

CASE 2 — Frisbee Clue: toy or sports equipment. Meaning: purpose changes **label**.

The case to investigate

CASE 3 — Wet slide Clue: equipment and possible hazard. Meaning: context matters.

The case to investigate

CASE 4 — Mystery object Clue: insufficient features. Meaning: need **more information**.

Keep this boundary clear

BOUNDARY **Label**s do not exist naturally in the data; people define them for a purpose.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Change B's quantity from 5 to 4; leave colour blank.

Predicted result

Prediction: B becomes review.

Why the result follows

Reason: The known failure is removed, but a required field is still missing.

Words that unlock the idea

> Use the vocabulary of Categories and Labels to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
category	group created for a purpose	
label	name assigned to an example	
overlap	when more than one label can fit	
ambiguity	when evidence allows multiple reasonable meanings	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR Every object has exactly one correct **label**.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Operational definition	A definition precise enough to tell someone what to observe or do when assigning a category
Annotator	A person who assigns labels to examples
Disagreement	A case where annotators assign different labels to the same example

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STUDENT LESSON · 6 OF 18

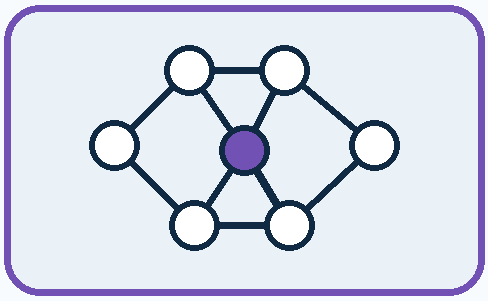
Follow a worked example

> Follow the worked trace for Categories and Labels, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A playground object can be both red and round, a toy and sports equipment, or safe in one situation and unsafe in another.

Step 1

STEP 1 — OBSERVE objects and a purpose for sorting

Step 2

STEP 2 — TRACE people define **label**s and assign examples

Step 3

STEP 3 — CONCLUDE **category** names attached to items

Model answer

MODEL REASONING Strong **label**s include a purpose and a rule for membership. **Overlap**ping objects may need multiple **label**s or a priority rule.

A revised definition should reduce **disagreement** without pretending every case is simple.

A claim to avoid

DO NOT OVERCLAIM **Label**s do not exist naturally in the data; people define them for a purpose.

What new **evidence** could change this answer?

A closer explanation

Card fields	Relevant evidence	Label and reason
Day 12 room A	Both values satisfy the criteria	Ready because both checks pass
Day 31 room A	31 is outside 1 through 30	Revise because a known value fails
Day 12 room unreadable	No known failure but the room is unknown	Review until the room is checked
Day 31 room unreadable	The day already fails although the room is unknown	Revise because the first check has priority

For the last card, first read only what is known: day 31. Compare it with the allowed range and record a failure. The guide therefore selects revise before the missing-room check.

Checking the room later could reveal another problem, but it cannot make day 31 satisfy this policy.

A clear explanation cites the particular criterion. Saying It looks wrong is not enough. Write Day 31 exceeds the accepted maximum of 30 instead.

That explanation lets another person reproduce the check and challenge it if the recorded value or policy is different.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Categories and Labels

A fictional craft club **label**s cardboard-order cards ready, review or revise.

The facts we will use

Supplied facts

Allowed quantity: whole number 1–4. Required colour: red or blue.

Supplied facts

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

AI ACADEMY / AI-1/7

Categories and Labels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read policy

Check known invalid fields before checking missing ones.



2. Card A

4 is allowed and red is allowed: ready.



3. Card B

5 is outside 1-4: revise, even though colour is missing.



4. Card C

2 is valid but colour is blank: review.

CHANGE ONE THING

Change B's quantity from 5 to 4; leave colour blank.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read policy	Check known invalid fields before checking missing ones.
2. Card A	4 is allowed and red is allowed: ready.
3. Card B	5 is outside 1–4: revise, even though colour is missing.
4. Card C	2 is valid but colour is blank: review.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure.

What this example does not establish

Keep the stated limit

Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

Keep the stated limit

Read the **label**led picture: Explain why B originally was revise rather than review, then **label** its changed version.

Keep the stated limit

Change one thing: Change B's quantity from 5 to 4; leave colour blank. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Categories and Labels claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Red ball	red and round	multiple descriptive labels
Frisbee	toy or sports equipment	purpose changes label
Wet slide	equipment and possible hazard	context matters
Mystery object	insufficient features	need more information

Repair two tempting claims

Challenge claim A

CLAIM A Every object has exactly one correct **label**.

Repair:

Challenge claim B

CLAIM B **Disagreement** proves one person was careless.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

If two **annotator**s agree on four of six cards, we know that four assigned **label**s matched. We do not yet know whether those **label**s followed the guide. Both **annotator**s might have made the same mistake.

Check their reasoning against the criteria and, where necessary, the source record.

Check the claim against the facts

Even perfect agreement under our calendar policy would not prove that the room is really booked. It would only show agreement about whether the printed fields pass the specified checks.

The policy itself may need changes for a real calendar. Record the policy version so examples **label**led under different instructions are not mixed without review.

Two people label every card ready. Explain why their agreement alone does not establish correct labels or confirm the events

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STUDENT LESSON · 8 OF 18

Practise together

Practise Categories and Labels with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Label**s are human choices that may **overlap** or be unclear.

Use a fresh example to explain **category**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: **Disagreement** proves one person was careless.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Mina **label**s day 30, room B as ready; Theo **label**s it revise because he thought the range stopped before 30. The **disagreement** points to a boundary interpretation. The guide says including both endpoints, so 30 passes.

The appropriate repair is to clarify and practise the inclusive boundary, not to change the **label** merely because one student is more confident.

> **Label day 0 with room A and day 18 with a blank room. Cite the exact rule for each and explain why the two cards need different next steps**

Ask **annotator**s to **label** examples independently before comparing. If they copy one another, agreement no longer tells us whether the guide can be applied independently.

Keep the original **label**s and reasons when discussing differences, then record any corrected **label**s separately.

Practise with less help: Replace a vague instruction

Use workbook W2 for the connected example “Categories and operational **label**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Explain why B originally was revise rather than review, then **label** its changed version.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

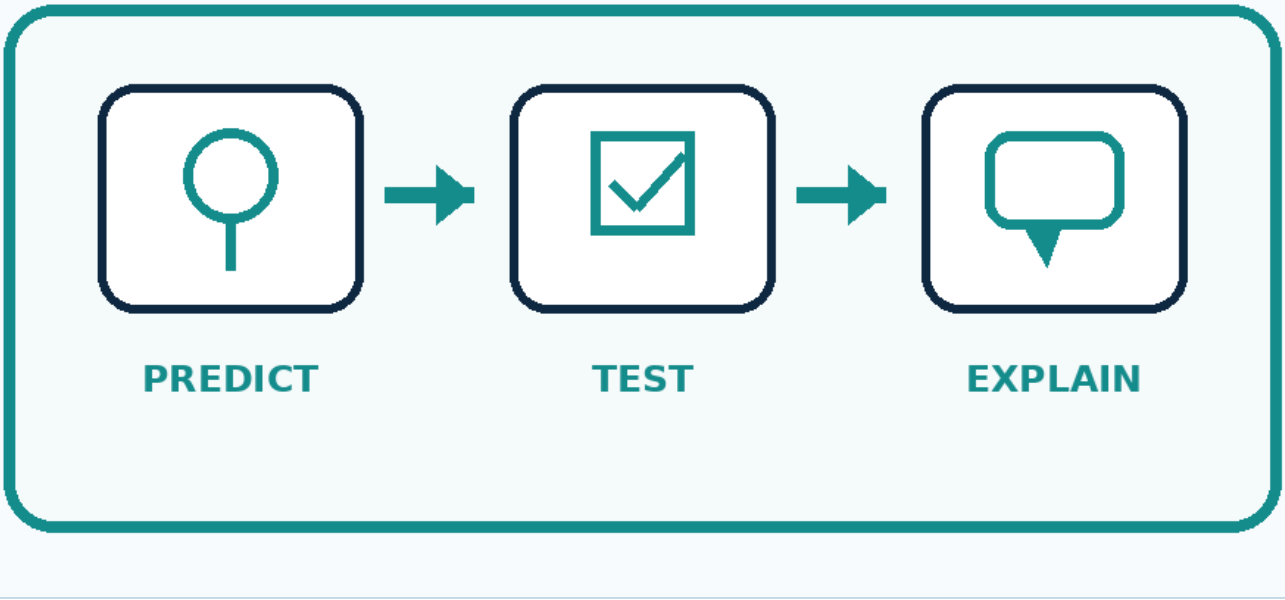
Run the investigation

> Investigate Categories and Labels. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Playground Label Council

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Design **label**s for playground objects; challenge ambiguous cards.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

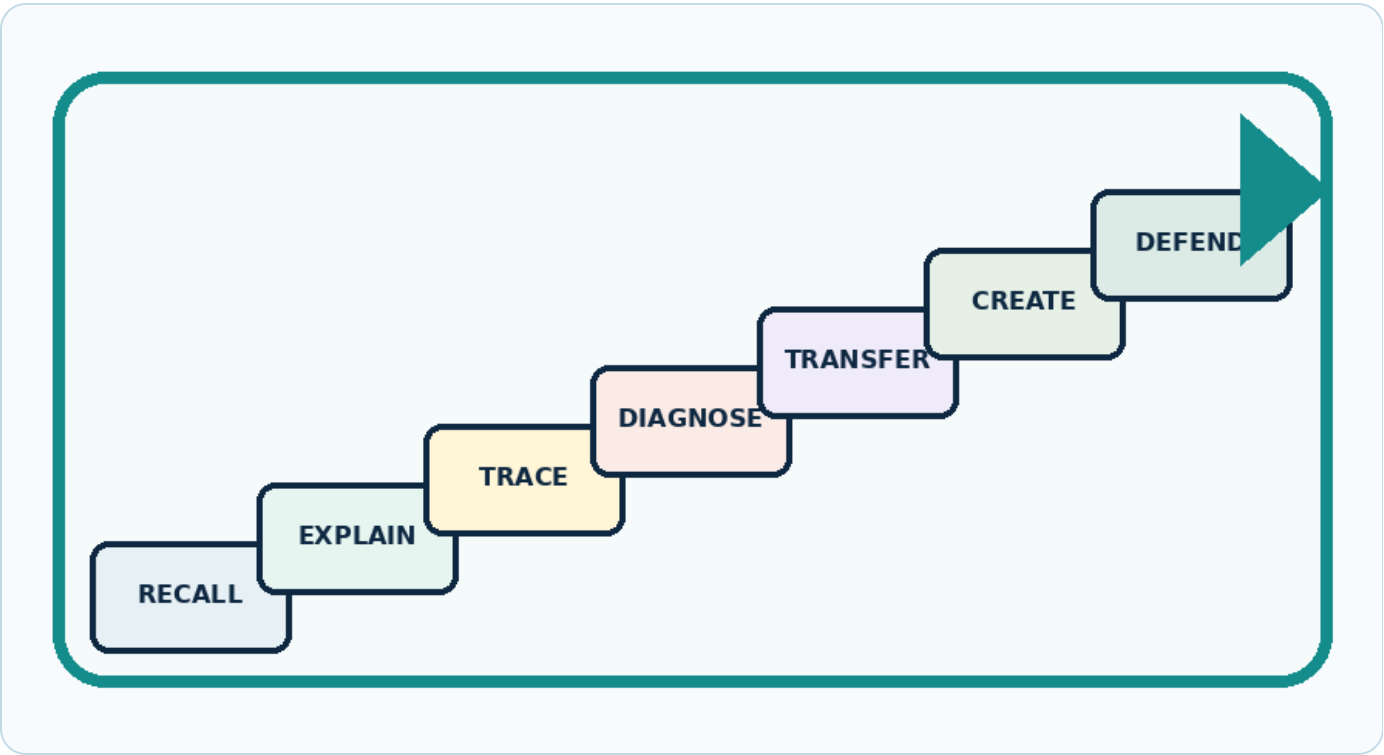
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Categories and Labels. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **category**.

EXPLAIN **Label**s are human choices that may **overlap** or be unclear.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Every object has exactly one correct **label**.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

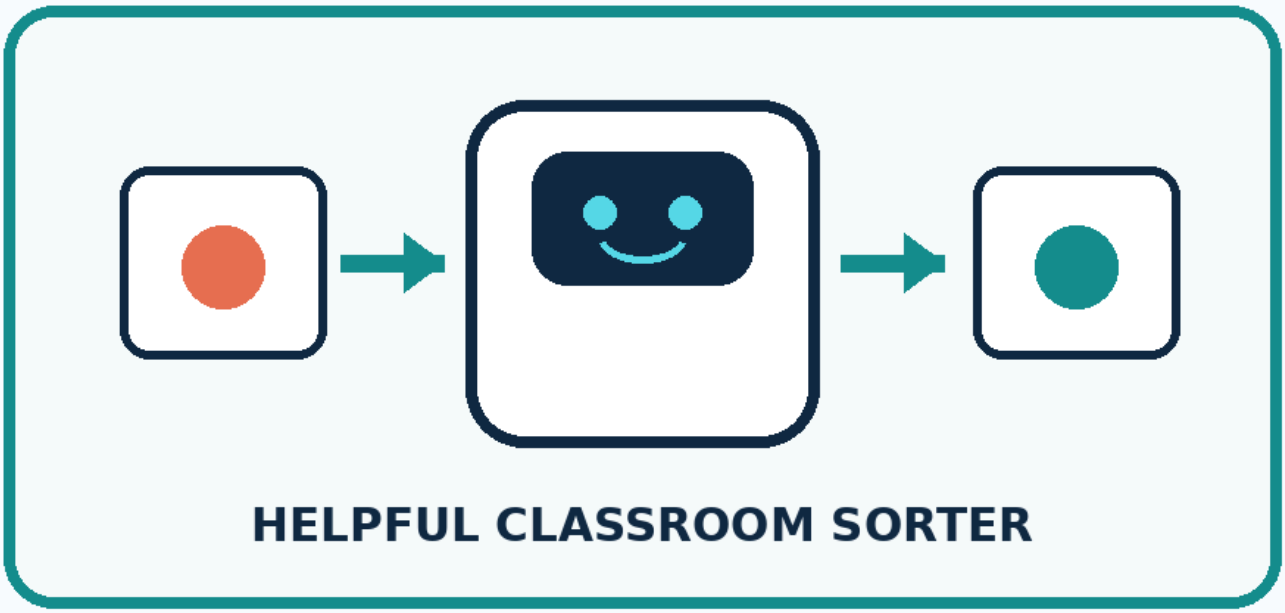
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Define the Classroom Sorter's bin **label**s and write how to handle objects that reasonably fit more than one bin.

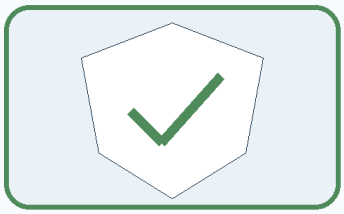
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN **Label**s do not exist naturally in the data; people define them for a purpose.

My project decision:

Evidence for your project

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Create **label**s for six playground objects, identify two **overlap**ping cases,

and revise one **label** definition so another team can apply it consistently.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Strong **label**s include a purpose and a rule for membership. **Overlap**ping objects may need multiple **label**s or a priority rule. A revised definition should reduce **disagreement** without pretending every case is simple.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Features: Useful Clues.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Categories and Labels case for supported practice. Repeated examples do not become fresh assessments.

Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Six cards show two circles, two triangles and two squares. Sort by shape.

The question

> Why can red and blue circles share a **label**?

Worked explanation

Both meet the circle definition. Colour does not change the agreed shape **label**. Apply one definition consistently.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Use the same label rule for every card

Evidence or stage	Value or result
Circles	2

Evidence or stage	Value or result
Triangles	2
Squares	2

Six cards show two circles, two triangles and two squares. Sort by shape.

> Why can red and blue circles share a **label**?

Both meet the circle definition. Colour does not change the agreed shape **label**. Apply one definition consistently.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Transfer the method to a new record

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For this toy sorter, label an object Seed-bearing when seeds are observed, Other when inspection establishes that it has no seeds, and Review when the needed contents cannot be observed. Sort an opened apple with visible seeds, a pencil and a sealed mystery box. Which item needs more **evidence**, and why?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Define the Classroom Sorter's bin **label**s and write how to handle objects that reasonably fit more than one bin.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Categories and Labels, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional craft club **label**s cardboard-order cards ready, review or revise.

By the end, I can	How I will show it
Apply a category definition using explicit evidence and boundary rules	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a known failure from information that is missing or unreadable	Explain the deciding step, show a trace or test, and state one limit.
Use disagreement to improve a labelling guide without mistaking agreement for truth	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Categories and Labels without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Apply a **category** definition using explicit **evidence** and boundary rules

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft club **label**s cardboard-order cards ready, review or revise.

Practice 1: explain the deciding evidence

Which **label**led step matches this detail? 2 is valid but colour is blank: review.

1. Read policy

2. Card B

3. Card A

4. Card C

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Card C 2 is valid but colour is blank: review. Operational **label**s specify what a word means for this task.

Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong. Revise is supported by a known failure.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change B's quantity from 5 to 4; leave colour blank. Which conclusion is supported by the supplied facts?

1. B becomes review.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: B becomes review. The known failure is removed, but a required field is still missing.

Repair a missing building block

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used **label**s and which stage used only the new measurement..

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

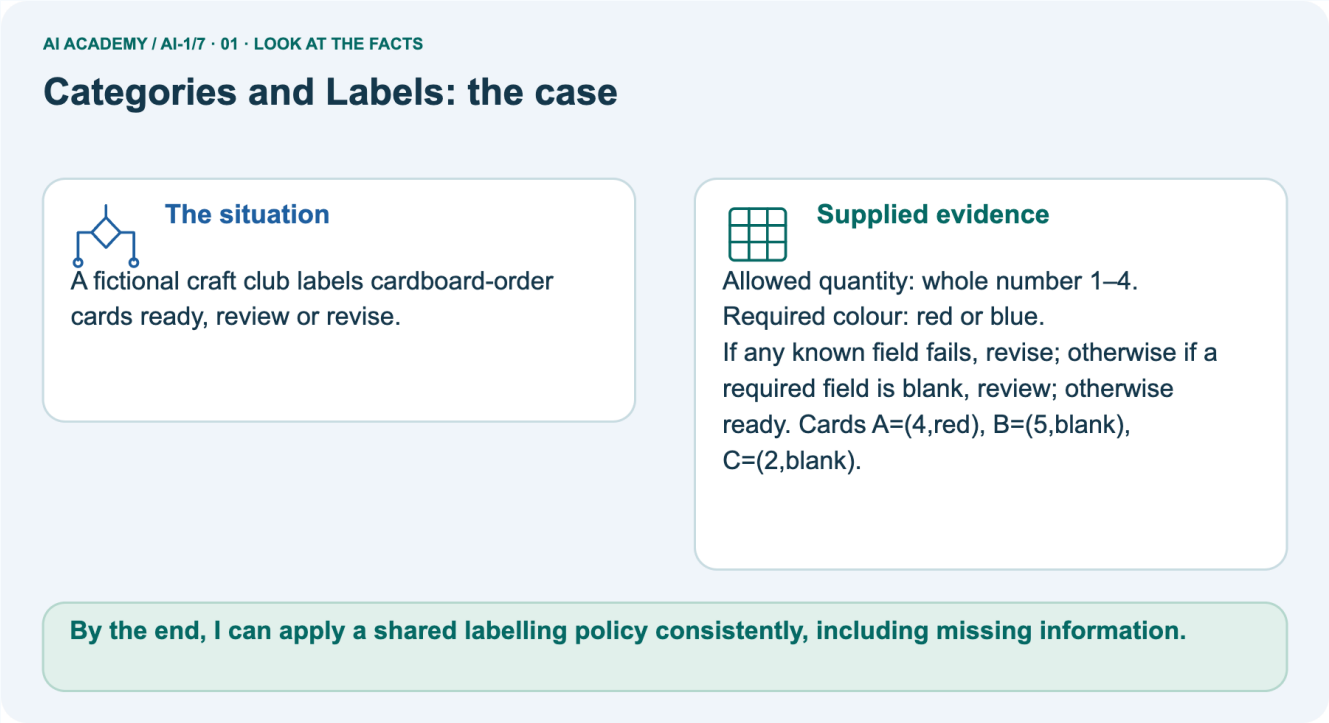
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Categories and Labels graphic using its labels, deciding evidence and relevant limitation.

01. Categories and Labels: the case



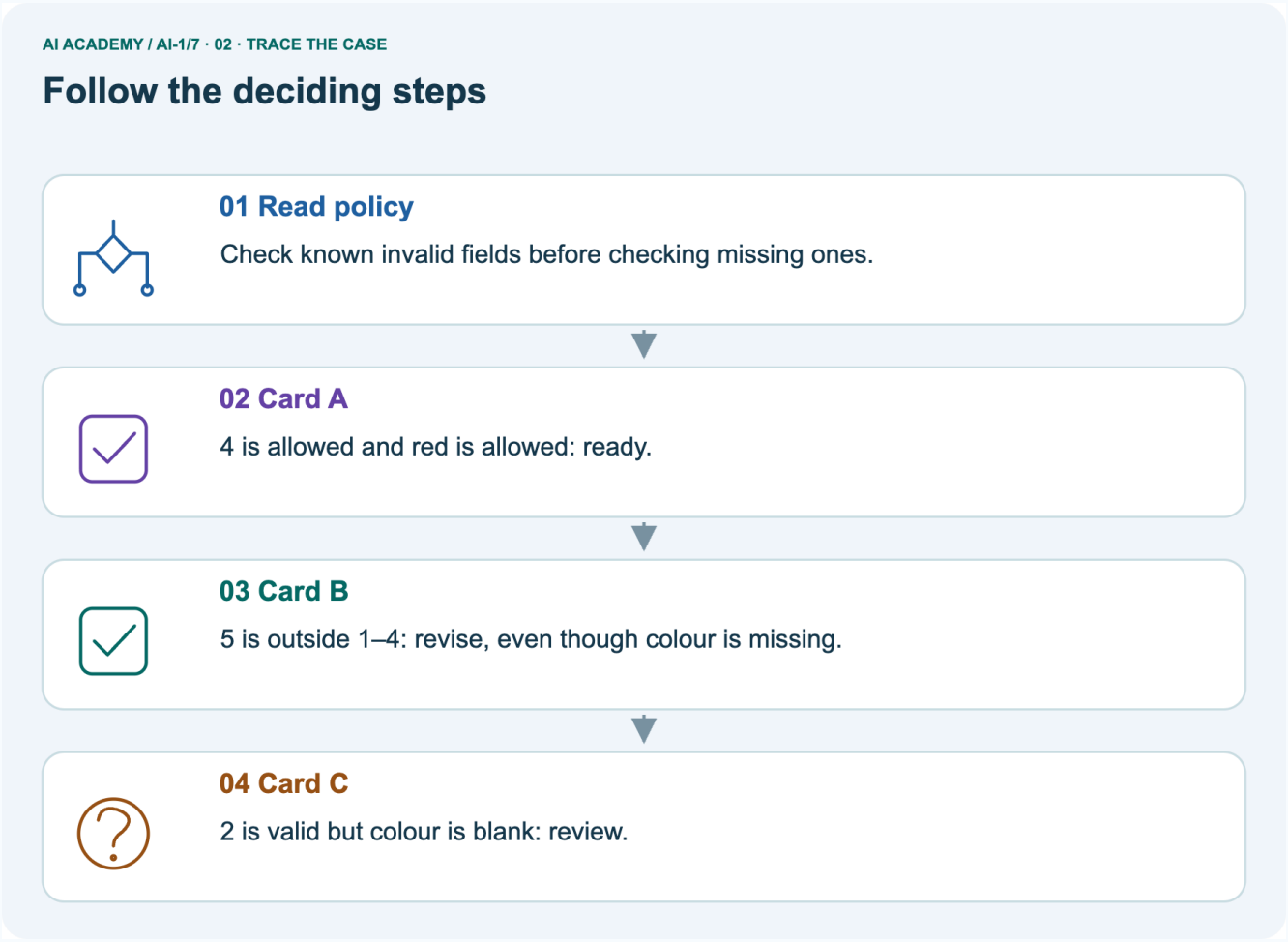
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft club **label**s cardboard-order cards ready, review or revise. Supplied facts: Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read policy: Check known invalid fields before checking missing ones. → Card A: 4 is allowed and red is allowed: ready. → Card B: 5 is outside 1–4: revise, even though colour is missing. → Card C: 2 is valid but colour is blank: review.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Three ordered checks for a calendar card

AI ACADEMY / AI-1/7 · 03 · INSPECT THE LESSON MODEL

Three ordered checks for a calendar card

Check in this order	Evidence	Recorded label
First check known values	Any known field fails its criterion	Revise even if another field is unknown
Then check missing information	No known failure but a field is blank or unreadable	Review because evidence is incomplete
Otherwise all checks have passed	Day is 1 through 30 and room is A or B	Ready under this record policy

Stop at the first matching row. The order resolves a card with both a known failure and an unknown field. Ready describes the printed record under this policy; it does not confirm that the event exists or has permission to run.

Follow the labelled picture with the matching explanation. Open the vector diagram

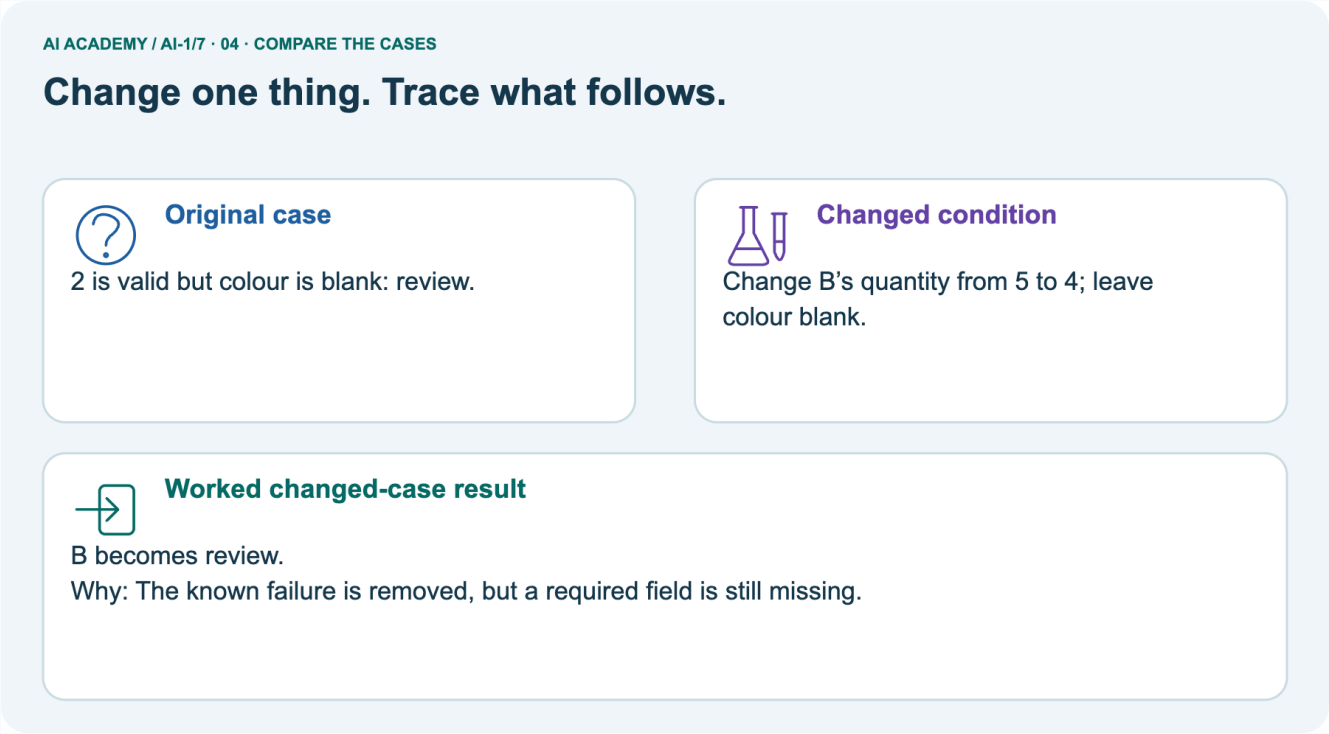
Three ordered checks for a calendar card. First check known values | Any known field fails its criterion | Revise even if another field is unknown.

Then check missing information | No known failure but a field is blank or unreadable | Review because **evidence** is incomplete.

Otherwise all checks have passed | Day is 1 through 30 and room is A or B | Ready under this record policy

> Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



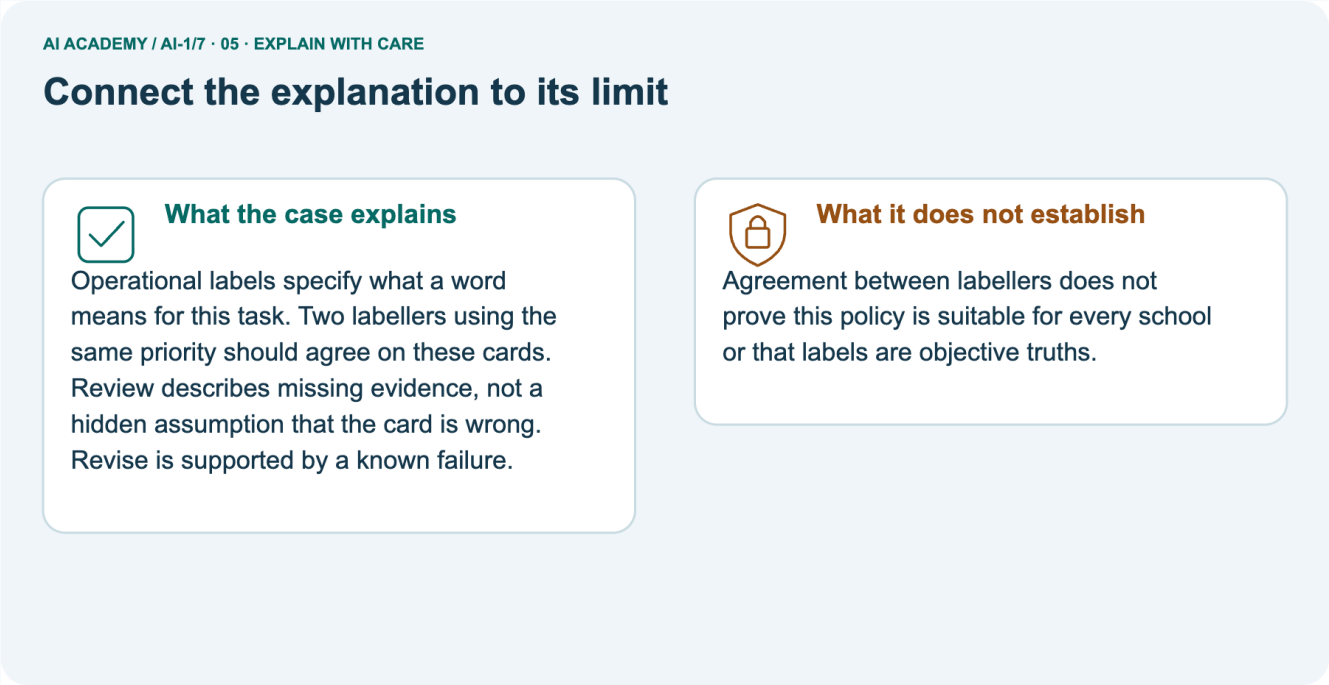
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 2 is valid but colour is blank: review. Changed condition: Change B's quantity from 5 to 4; leave colour blank. Expected result: B becomes review.

Reason: The known failure is removed, but a required field is still missing.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



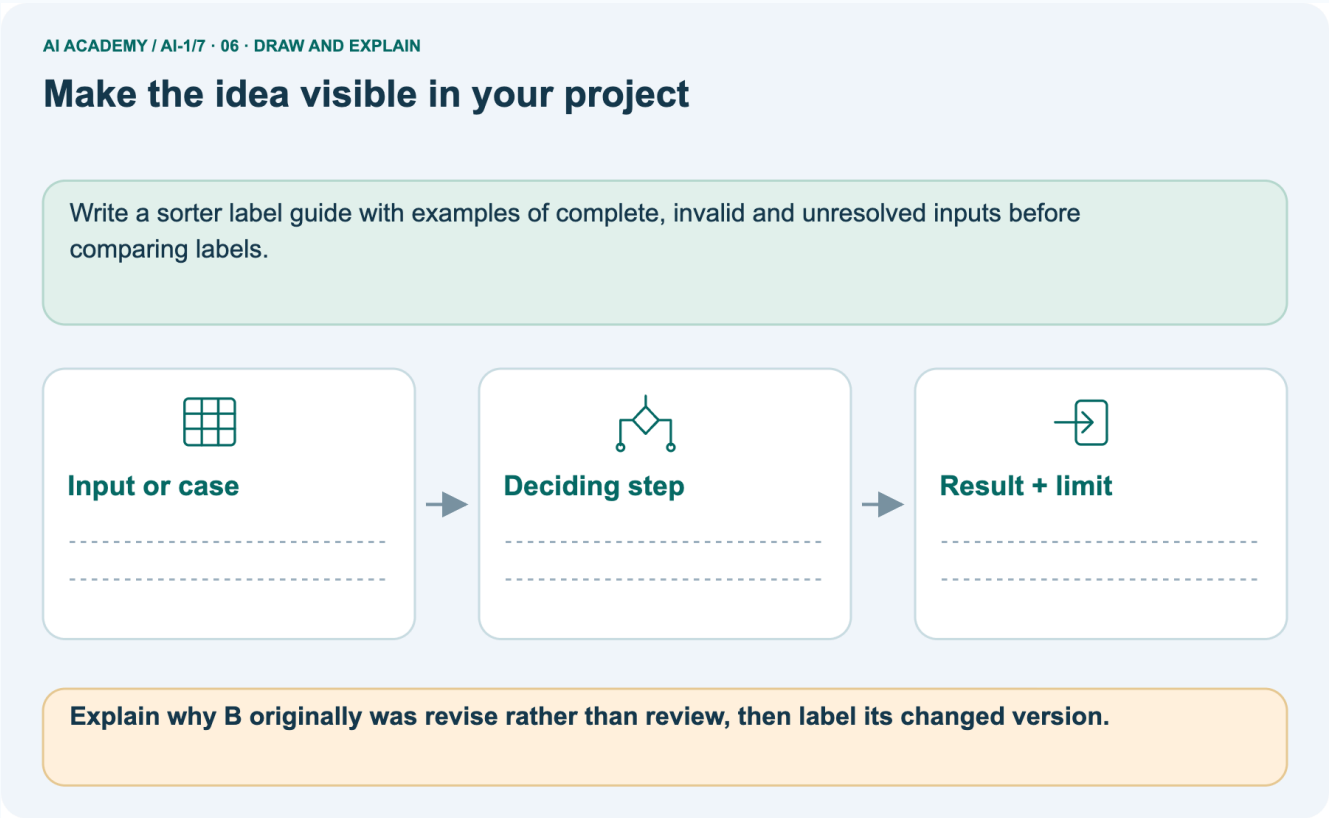
Follow the labelled picture with the matching explanation. Open the vector diagram

Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure. Limit: Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B originally was revise rather than review, then **label** its changed version.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.